A Frequency-Guided Axial Refinement Network for High-Resolution Remote Sensing Image Semantic Segmentation

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***Abstract*—The remote sensing image semantic segmentation is fundamental to fine-grained land-cover interpretation. However, in high-resolution complex scenes, segmentation performance is still limited by strong interclass similarity, large-scale variation, and the frequent degradation of boundary details. To address these issues, we propose a frequency-guided axial refinement network (FGAR-Net). The FGAR-Net adopts an asymmetric stepwise encoder–decoder architecture. Specifically, a frequency-guided channel recalibration (FGCR) module introduces discrete cosine transform (DCT) priors into skip pathways to suppress background redundancy and highlight discriminative cues. At the bottleneck stage, an axial-window perception (AWP) module jointly captures regional semantics, long-range structural dependencies, and local details. During decoding, the adaptive weighted fusion and context refinement (AWCR) blocks coordinate deep and shallow features to reduce cross-layer semantic inconsistency, while a final context-guided feature refinement (CFR) module imposes global contextual constraints to accurately restore local high-resolution details. Extensive experiments demonstrate that FGAR-Net achieves mean intersection over unions (mIoUs) of 85.17%, 87.55%, and 65.75% on the International Society for Photogrammetry and Remote Sensing (ISPRS) Vaihingen, ISPRS Potsdam, and Wuhan high-resolution dense labeling dataset (WHDLD) datasets. These results indicate that FGAR-Net provides an effective solution for fine-grained segmentation in complex remote sensing scenes.**

***Index Terms*—Context-guided refinement, cross-level feature fusion, frequency-domain feature modeling, high-resolution imagery, remote sensing semantic segmentation.**

I. Introduction

THE remote sensing observation plays a crucial role in environmental monitoring, urban management, resource investigation, and disaster assessment. As a pixel-level parsing task, the remote sensing image semantic segmentation assigns a semantic label to each pixel. It serves as a fundamental basis for fine-grained land-cover recognition and complex scene understanding. With the development of deep learning, this field has made continuous progress in feature representation, context modeling, and dense prediction. However, achieving accurate, continuous, and structurally consistent semantic parsing in high-resolution complex scenes remains a challenge.

Compared with natural images, the high-resolution remote sensing imagery (HRRSI) presents more complex spatial layouts, dramatic scale variations, and denser object distributions [1]. In these scenes, different categories often exhibit high visual similarity. Meanwhile, the samples within the same category can show large intraclass variations due to texture, illumination, shadows, or background interference. Furthermore, fine regions, such as road boundaries, building contours, small objects, and thin structures, easily degrade during repeated downsampling and feature propagation [2]. More importantly, these difficulties are often amplified during decoding. Deep features provide strong semantic discrimination but weak responses to local details and boundaries. Conversely, the shallow features preserve rich spatial textures and structural information but are highly vulnerable to background redundancy and local noise [3]. Consequently, as illustrated in Fig. 1(b), direct cross-layer addition or concatenation

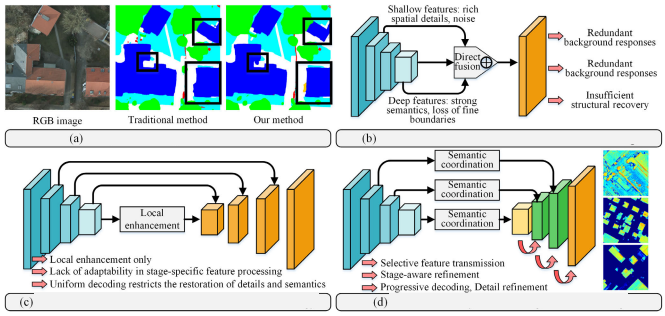


Fig. 1. Motivation of the proposed method. (a) Visual comparison of segmentation results in a complex high-resolution scene. (b) Limitations of conventional direct fusion decoding, which often suffers from background redundancy and insufficient structural recovery. (c) Limitations of uniform local enhancement strategies that lack stage-aware adaptability. (d) Proposed asymmetric progressive decoding paradigm, which integrates selective feature transmission and stage-aware refinement for accurate parsing.

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often fails to simultaneously preserve semantic consistency, structural continuity, and high-resolution details.

The existing studies have improved remote sensing semantic segmentation from multiple aspects. To alleviate category ambiguity, researchers have introduced knowledge graphs [4], vision-language associations [5], and domain generalization strategies [6] to improve representation robustness. For feature fusion and alignment, multimodal paradigms [7], sparse transformer fusion [8], interactive alignment [9], and decoupled interaction designs [10] have been proposed to reduce semantic mismatch across layers. To enhance contextual modeling and boundary recovery, the previous methods explore compact global modeling [11], multiscale aggregation [12], [13], hybrid CNN–transformer architectures [14], [15], [16], generative refinement [17], and denoising-based localization [18]. Although these studies improve scene parsing, most focus on strengthening individual components [see Fig. 1(c)]. They lack a unified design for the decoding process, particularly a stagewise recovery mechanism that addresses the distinct objectives of different decoding stages [19]. This limitation is highly evident in specialized extraction tasks. For instance, the HRRSI extraction requires not only long-range contextual reasoning but also the effective suppression of shallow noise in skip pathways [20]. Unconstrained shallow priors can easily introduce redundant responses. To improve regional completeness and boundary continuity, consistency learning [21] and self-supervised alignment [22] offer valuable insights. Together, these studies indicate that a uniform fusion strategy is insufficient for distinct recovery demands. Specifically, feature delivery requires noise suppression; the bottleneck stage requires coordinated modeling of global semantics and local structures; and the intermediate and final decoding stages emphasize semantic coordination and detail reconstruction, respectively.

Based on these observations, we argue that improving HRRSI semantic segmentation relies on building a decoding pipeline that explicitly integrates selective feature delivery, structural modeling, and stage-aware recovery, rather than simply stacking isolated modules. To this end, we propose the frequency-guided axial refinement network (FGAR-Net). As shown in Fig. 1(d), unlike conventional symmetric frameworks, the FGAR-Net adopts an asymmetric stepwise encoder–decoder architecture. It organizes hierarchical feature extraction, selective skip-feature delivery, bottleneck structural modeling, and stage-aware decoding refinement into a unified pipeline. Specifically, along skip pathways, we introduce a frequency-guided channel recalibration (FGCR) module. It uses discrete cosine transform (DCT) priors to drive channel recalibration, suppressing background redundancy before feature delivery and strengthening discriminative cues for confusing categories. At the bottleneck stage, an axial-window perception (AWP) module decouples spatial perception into regional, axial, and local branches. This enables the coordinated modeling of regional semantics, long-range structural dependencies, and local detail responses. During decoding, we adopt a stage-aware recovery strategy. In intermediate stages, the adaptive weighted fusion and context refinement (AWCR) module coordinates deep and shallow features to alleviate cross-layer semantic mismatch. In the final high-resolution stage, the context-guided feature refinement (CFR) module imposes global contextual constraints and local fine-grained calibration to improve the recovery of complex boundaries and small targets, ultimately yielding more accurate and structurally complete parsing results [Fig. 1(a)].

The main contributions of this work are summarized as follows.

1) We propose FGAR-Net, an asymmetric stepwise encoder–decoder network for high-resolution remote sensing image semantic segmentation. It explicitly unifies selective feature delivery, structural modeling, and stage-aware recovery into a single pipeline.
2) We design the FGCR module and the AWP bottleneck. The FGCR module suppresses background redundancy and strengthens discriminative cues in skip pathways, while the AWP module jointly models regional semantics, long-range continuous structures, and local details.
3) We develop a stage-aware decoding refinement scheme. It utilizes AWCR blocks for cross-layer semantic coordination during intermediate decoding and a CFR module for local boundary recovery in the final high-resolution stage.
4) Extensive experiments on the International Society for Photogrammetry and Remote Sensing (ISPRS) Vaihingen, ISPRS Potsdam, and Wuhan high-resolution dense labeling dataset (WHDLD) datasets demonstrate the effectiveness of FGAR-Net, verifying its applicability to fine-grained semantic segmentation in complex high-resolution remote sensing scenes.

II. Related Work

A. Semantic Segmentation of High-Resolution Remote Sensing Images

The semantic segmentation of remote sensing images focuses on pixel-level classification in complex surface scenarios. Early CNN-based research heavily relied on dual-branch structures, feature pyramids, and multiscale attention mechanisms (e.g., ABCNet [23], A^2-FPN [24], CERNet [25], multiscale and global information modeling network [26], and HMA-UNet [27]) to enhance local spatial feature representation. Constrained by the local receptive field of convolutions, such methods struggle to establish long-range semantic associations. With the introduction of transformer and hybrid architectures (e.g., MFNet [28], IGECNet [29], and FTUNetformer [30]), the superiority of self-attention mechanisms in global spatial relationship modeling has been demonstrated, thereby improving the parsing capability for large-scale scenes. Meanwhile, targeting the unique characteristics of sparse annotations and long-tail distributions in remote sensing scenes, the approaches, such as weakly and semi-supervised learning [31], multisource self-supervised fusion [32], open-vocabulary segmentation [33], scribble-guided cross-attention [34], and joint networks for hyperspectral graph structures [35], have broadened the application boundaries from different dimensions.

Recently, the state-space models, particularly Mamba-based architectures, have emerged as an efficient alternative for high-resolution remote sensing image segmentation.

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The representative methods, such as MF-Mamba [36], ReSMamba [37], and MambaRSIS [38], exploit selective state-space modeling to capture long-range dependencies with linear computational complexity. The subsequent studies have further extended Mamba toward heterogeneous architectural collaboration, multimodal interaction, multiscale fusion, and frequency-domain modeling. The MCPNet [39] adopts an asymmetric Mamba–CNN architecture, coupling a deep Mamba branch for global dependency modeling with a detail-preserving CNN branch for fine spatial representation. Its multigated fusion module progressively aligns heterogeneous features, balancing segmentation accuracy and memory efficiency. For multimodal segmentation, MGF-GCN [40] employs modality-specific encoders and MHIMamba to integrate complementary red-green-blue/infrared-red-green (RGB/IRRG) and digital surface model (DSM) information, followed by a progressive decoder for detail recovery. The PMFM [41] incorporates frequency–phase cues into positional encoding and combines cross-attention with Vision Mamba for multiscale frequency–spatial fusion. The WCI-Mamba [42] further couples wavelet decomposition with a cross-frequency state-space model, jointly modeling high- and low-frequency components to enhance structural consistency and boundary recovery under intensity inhomogeneity.

These studies demonstrate the flexibility of Mamba in global dependency modeling, heterogeneous feature collaboration, multimodal fusion, and frequency-aware representation. However, most existing designs focus on enhancing a specific representation dimension or constructing task-oriented fusion mechanisms. They do not explicitly distinguish the heterogeneous recovery objectives at different decoding depths. In particular, the shallow skip pathways require effective suppression of background redundancy, the bottleneck stage requires coordinated modeling of regional semantics and continuous structures, the intermediate decoding requires cross-level semantic alignment, and the final high-resolution stage emphasizes boundary and small-object reconstruction. Therefore, the efficient long-range modeling alone remains insufficient for fine-grained remote sensing scene parsing, motivating a stage-aware decoding framework that jointly considers selective feature delivery, structural perception, semantic coordination, and high-resolution detail refinement.

B. Context Modeling and Multiscale Structural Perception

The scale variation of remote sensing surface objects and interclass texture homogenization requires pixel-level prediction to integrate cross-scale structures and contextual relationships. One conventional paradigm focuses on global–local joint modeling, such as MSLK-EAN with large-kernel attention [39], global context iterative refinement [40], and mutual attention CMANet [41], leveraging spatial interaction to suppress local semantic ambiguity. FSSFFormer [42] further validates the necessity of multiscale synergy in the frequency domain. Another paradigm focuses on spatial topological relationships, utilizing class-guided interactions [43], dataset- and image-level interoperability [44], hierarchical perception modules [45], and topological curve constraints [46] to reinforce object edges and continuous features. Similar design logic appears in adjacent dense prediction tasks, such as scalewise bidirectional alignment in referring segmentation [47], cross-layer interaction in RGB-T fusion [48], and bidirectional cross-attention in saliency detection [49]. Furthermore, the introduction of external priors, such as cross-image consistency constraints, controllable diffusion data priors, and spatial attention module (SAM) boundary assistance, confirms the guiding role of the global knowledge domain in local parsing.

The existing context enhancement mechanisms face an inherent structural tradeoff: dense global attention tends to smooth features, easily diluting local high-frequency responses, whereas conventional multiscale downsampling operations easily disrupt the spatial topology of slender ground objects. Breaking this bottleneck depends on constructing a multidimensional perception mechanism, establishing long-range axial dependencies to maintain macrotopological continuity, while precisely aggregating local window semantics to preserve microfine-grained features, thereby achieving the synergy of structure and detail.

C. Cross-Level Feature Fusion and Decoder Refinement

The upper bound of the final segmentation quality is simultaneously constrained by cross-layer fusion and the decoding paradigm [50]. Deep features possess strong semantic priors but limited spatial resolution, while shallow features contain fine textures interlaced with massive background noise. The conventional methods mostly adopt skip connections or basic interaction modules to bridge hierarchical structures. Extensive research has shown that direct cross-layer isomorphic transmission indiscriminately amplifies feature noise, uncertainty, and semantic misalignment across layers, exacerbating boundary blurriness and detail distortion in the final prediction [51], [52].

To address the defects of isomorphic fusion, the design of the decoding stage has progressively evolved toward selective alignment. MambaRSIS [38] introduces a context-guided aggregation mechanism to alleviate multiscale semantic mismatch, while some studies utilize bidirectional cross-attention to supplement adjacent layer semantics. Centered on boundary recovery, FreDNet [53] and PV Segmenter [54] introduce frequency-domain priors into cross-stage fusion, relying on high-frequency components to separate structures from noise; the frequency-domain context decoding in IGECNet [29] also corroborates the effectiveness of this dimension. A universal limitation of the existing decoding architectures lies in the homogenization of the transmission mechanism, lacking stagewise adaptation for different decoding depths: shallow fusion focuses on noise suppression, whereas deep fusion targets semantic calibration. Indiscriminate feature transmission hinders optimal decoding. This inherent conflict motivates the design requirement for a stage-sensitive decoding system, namely, utilizing frequency-domain priors to implement channel-level high-frequency filtering during cross-layer transmission, combined with context priors for adaptive multistage feature reorganization and refinement.

III. Methods

The HRRSI segmentation remains challenged by high interclass similarity, large-scale variation, and the frequent loss of fine boundary structures in complex scenes. To address these

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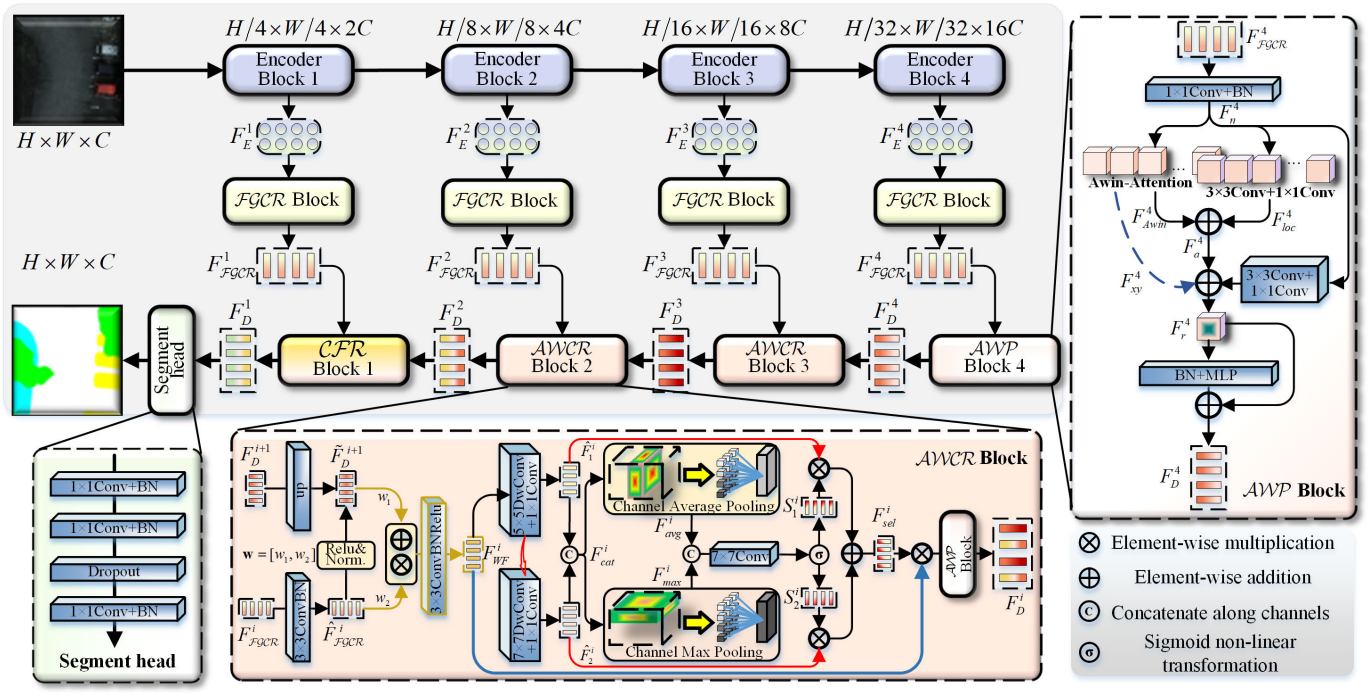


Fig. 2. Overview of the FGAR-Net.

issues, we propose the FGAR-Net (as show in Fig. 2). Unlike conventional symmetric encoder–decoder frameworks, FGAR-Net adopts an asymmetric stepwise encoding–decoding design, in which hierarchical representation learning, selective feature delivery, and progressive feature refinement are organized into a unified segmentation pipeline.

Given an input image $I \in \mathbb{R}^{H\times W\times C}$, the hierarchical encoder progressively extracts multilevel features $F_E^i \in \mathbb{R}^{H/2^i\times W/2^i\times C}$, $i \in \{1,2,3,4\}$. To recalibrate the encoded representations and prevent the transmission of background redundancy, the FGCR blocks are applied at each skip pathway, producing enhanced frequency-guided feature maps F_{FGCR}^i.

The decoding process is stage-aware rather than adopting a uniform refinement strategy. At the bottleneck stage ($i = 4$), the AWP module processes F_{FGCR}^4 to model long-range dependencies, yielding the initial decoded feature F_D^4. For intermediate decoding stages ($i \in \{2,3\}$), the AWCR modules integrate the upsampled deep features F_D^{i+1} with the corresponding F_{FGCR}^i, producing F_D^3 and F_D^2. At the final high-resolution stage ($i = 1$), the CFR module refines the fused representation to yield F_D^1, which is then fed into the segmentation head to generate the final prediction.

A. FGCR Module

In HRRSI segmentation, the shallow features extracted from the encoder via skip connections encode rich spatial details, but they are often heavily corrupted by high-frequency background noise and intraclass variance. Relying solely on spatial-domain convolutions is insufficient to suppress such noise or highlight discriminative structures. To address this issue, we propose the FGCR module (Fig. 3, Algorithm 1),

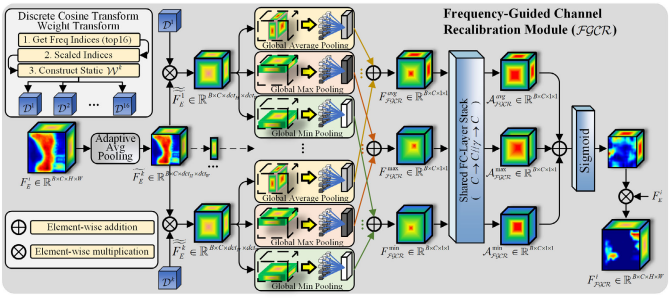


Fig. 3. Overview of the FGCR module.

which introduces DCT priors to dynamically reweight channel importance before these features are delivered to the decoder.

Given the intermediate encoder feature $F_E^i \in \mathbb{R}^{B\times C\times H\times W}$, the module first selects the top-K frequency indices to construct a set of fixed 2-D DCT basis weights $\mathcal{D}^k \in \mathbb{R}^{d_h^i\times d_w^i}$, where $k \in \{1,2,\ldots,K\}$. To align the spatial resolution with the predefined frequency scale, adaptive average pooling is applied to F_E^i, yielding the scaled feature $\widetilde{F}_E^i \in \mathbb{R}^{B\times C\times d_h^i\times d_w^i}$. The frequency-aware spectral response for each branch is then explicitly generated through elementwise multiplication

$$\widetilde{\widetilde{F}}_E^k = \mathcal{D}^k \otimes \widetilde{F}_E^i. \tag{1}$$

To comprehensively capture the spectral distribution across diverse semantic regions, we extract three distinct spatial statistics from $\widetilde{\widetilde{F}}_E^k$: global average pooling (GAvgP), global max pooling (GMaxP), and global min pooling (GMinP). These statistics are accumulated across all K branches and normalized by K to form holistic frequency descriptors: F_{FGCR}^{avg}, F_{FGCR}^{max}, and $F_{\text{FGCR}}^{min} \in \mathbb{R}^{B\times C\times 1\times 1}$. Utilizing max and min

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pooling alongside average pooling is crucial for HRRSI: max pooling captures sharp boundary peaks, while min pooling retains sensitivity to homogeneous or shadowed regions.

Algorithm 1 Pseudocode of FGCR

Input: Encoder feature $F_E^i \in \mathbb{R}^{B\times C\times H\times W}$, $i \in \{1,2,3,4\}$; Discrete Cosine Transform size dCT_h, dCT_w, reduction ratio γ, frequency branches $k \in \{1,2,\ldots,K\}$
Output: Enhanced features $F_{\text{FGCR}}^i \in \mathbb{R}^{B\times C\times H\times W}$

$\{(h_k, w_k)\}_{k=1}^K \leftarrow$ *get frequency indices (top K)*
Scale (h_k, w_k) to the target DCT size (dCT_h, dCT_w)
Construct fixed DCT weights $\mathcal{D}^{k,h_k,w_k} \in \mathbb{R}^{dCT_h\times dCT_w}$ for all k
$\widetilde{F}_E^i \leftarrow AdaptiveAvgPool(F_E^i, (dCT_h, dCT_w))$
Initialize $F_{\text{FGCR}}^{avg} \leftarrow 0$, $F_{\text{FGCR}}^{max} \leftarrow 0$, $F_{\text{FGCR}}^{min} \leftarrow 0$
for $k = 1,\ldots,K$ **do**
$\widetilde{\widetilde{F}}_E^k \leftarrow \mathcal{D}^k \otimes \widetilde{F}_E^i$ ▷ Element-wise spectral response
*/*Accumulate three statistics (implementation)*/*
$F_{\text{FGCR}}^{avg} \leftarrow F_{\text{FGCR}}^{avg} + GAvgP(\widetilde{\widetilde{F}}_E^k)$
$F_{\text{FGCR}}^{max} \leftarrow F_{\text{FGCR}}^{max} + GMaxP(\widetilde{\widetilde{F}}_E^k)$
$F_{\text{FGCR}}^{min} \leftarrow F_{\text{FGCR}}^{min} + GMinP(\widetilde{\widetilde{F}}_E^k)$
end for
$F_{\text{FGCR}}^{avg} \leftarrow F_{\text{FGCR}}^{avg}/K$ ▷ Normalize by K
$F_{\text{FGCR}}^{max} \leftarrow F_{\text{FGCR}}^{max}/K$ ▷ Normalize by K
$F_{\text{FGCR}}^{min} \leftarrow F_{\text{FGCR}}^{min}/K$ ▷ Normalize by K
*/*Shared 2-layer 1×1 Conv (FC) with reduction γ*/*
$FC(\cdot) = Conv_{1\times1}(C \to C/\gamma) \to ReLU \to Conv_{1\times1}(C/\gamma \to C)$
$A_{\text{FGCR}}^{avg} \leftarrow FC(F_{\text{FGCR}}^{avg})$, $A_{\text{FGCR}}^{max} \leftarrow FC(F_{\text{FGCR}}^{max})$, $A_{\text{FGCR}}^{min} \leftarrow FC(F_{\text{FGCR}}^{min})$
$F_{\text{FGCR}}^i \leftarrow F_E^i \otimes \sigma(A_{\text{FGCR}}^{avg} + A_{\text{FGCR}}^{max} + A_{\text{FGCR}}^{min})$
Return F_{FGCR}^i

Subsequently, these three descriptors are fed into a shared fully connected network stack (comprising two 1×1 convolutions with a reduction ratio γ and a rectified linear unit (ReLU) activation) to model cross-channel dependencies, producing specific attention maps A_{FGCR}^{avg}, A_{FGCR}^{max}, and A_{FGCR}^{min}. Finally, the refined feature F_{FGCR}^i is obtained by aggregating these attention mappings, applying a sigmoid activation σ, and modulating the original spatial feature

$$F_{\text{FGCR}}^i = F_E^i \otimes \sigma\left(A_{\text{FGCR}}^{avg} + A_{\text{FGCR}}^{max} + A_{\text{FGCR}}^{min}\right). \tag{2}$$

By explicitly coupling multibranch frequency priors with channel attention, the FGCR module effectively suppresses irrelevant background noise and enhances the discriminative power for visually similar categories without relying on computationally expensive spatial self-attention.

B. AWP Module

The core mechanism of AWP (Algorithm 2) relies on a parallel multibranch architecture that decouples spatial perception into regional, axial, and local representations. After an initial projection and normalization, the feature F_n^4 undergoes three complementary transformations.

Algorithm 2 Pseudocode of AWP

Input: The input feature map $F_{\text{FGCR}}^4 \in \mathbb{R}^{B\times C\times H\times W}$, window size $w_s = 8$, number of attention heads $N_h = 16$, per-head channel dimension $d = C/N_h$, relative position bias $B_{\text{rel}} \in \mathbb{R}^{N_h\times(w_s^2)\times(w_s^2)}$
Output: Enhanced features $F_D^4 \in \mathbb{R}^{B\times C\times H\times W}$

$F_n^4 \leftarrow ConvBN_{1\times1}(F_{\text{FGCR}}^4)$ ▷ Input projection and feature normalization
$\widetilde{F}_n^4 \in \mathbb{R}^{B\times C\times H_p\times W_p}$, where $H_p = \lceil \frac{H}{w_s} \rceil w_s$ and $W_p = \lceil \frac{W}{w_s} \rceil w_s$ ▷ Pad F_n^4
*/*Partition Q, K, V into non-overlapping windows of size $w_s \times w_s$, and reshape*/*
$Q, K, V \in \mathbb{R}^{B_w\times N_h\times w_s^2\times d} \leftarrow Conv_{1\times1}(\widetilde{F}_n^4)$, where $B_w = B \times \frac{H_p}{w_s} \times \frac{W_p}{w_s}$
*/*Reverse the windows and recover the spatial layout*/*
$F_{\text{Awin}}^4 \in \mathbb{R}^{B\times C\times H_p\times W_p} \leftarrow \left(Softmax\left(\frac{Q(K)^T}{\sqrt{d}} + B_{\text{rel}}\right)\right)V$ ▷ Awin-Attention structure
$F_{\text{Awin}}^4 \leftarrow F_{\text{Awin}}^4[:,:,:H,:W]$ ▷ Crop to the original size
$F_{xy}^4 \leftarrow AvgPool_{(w_s,1)}(F_{\text{Awin}}^4) + AvgPool_{(1,w_s)}(F_{\text{Awin}}^4)$ ▷ Axial perception branch
$F_{\text{loc}}^4 \leftarrow Conv_{3\times3}(Conv_{1\times1}(F_n^4))$ ▷ Local perception branch
$F_r^4 \leftarrow F_{\text{Awin}}^4 + F_{xy}^4 + Conv_{3\times3}(Conv_{1\times1}(F_n^4)) + F_{\text{loc}}^4$ ▷ Feature aggregation and residual update
$F_D^4 \leftarrow F_r^4 + MLP(BN(F_r^4))$ ▷ MLP refinement
Return F_D^4

Regional Semantic Branch (Awin-Attention F_{Awin}^4): To capture regional contexts efficiently, F_n^4 is partitioned into nonoverlapping windows of size $w_s \times w_s$. Queries (Q), keys (K), and values (V) are derived, and a multihead self-attention mechanism is applied. To preserve spatial awareness within the windows, a trainable and flexible relative position bias B_{rel} is incorporated.

Long-Range Structural Branch (Axial Perception F_{xy}^4): The standard window attention isolates cross-window interactions. To bridge this gap and model continuous structural dependencies, which are crucial for linear HRRSI features, we apply horizontal and vertical axial average pooling to F_{Awin}^4.

Local Detail Branch (F_{loc}^4): In parallel, to prevent the smoothing out of fine-grained spatial details caused by attention mechanisms, a dedicated convolution pathway extracts local features.

Ultimately, a batch normalization and multilayer perceptron block refines the representations. By integrating these parallel spatial mechanisms, the AWP module successfully breaks the isolation of standard window attention. It strengthens the structural integrity of large and continuous targets via axial pooling, retains local boundary precision via convolutions, and establishes a robust, spatially aware global semantic anchor for the subsequent decoding stages. Mechanistically, the window attention establishes content-adaptive interactions within compact regions but models cross-window dependencies only indirectly. The AWP alleviates this limitation by applying horizontal and vertical axial pooling to the window-enhanced representation. Because these operations are performed at the low-resolution bottleneck, they cover a comparatively

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broad region in the original image and facilitate information aggregation among directionally coherent structures without dense full-map pairwise attention. Meanwhile, the parallel convolutional branch preserves local gradients and boundary responses that may otherwise be attenuated by contextual aggregation. The AWP therefore integrates regional semantics, directional structural continuity, and local details within a unified 2-D operator.

This design shares with Mamba-based state-space models the objective of extending the effective receptive field while avoiding dense global self-attention, but the underlying mechanisms differ. Mamba establishes long-range dependencies through selective state propagation along serialized scanning paths, providing content-adaptive sequence modeling with linear complexity. In contrast, the AWP does not serialize the 2-D feature map or perform state-space recurrence. Instead, it introduces explicit geometric priors through orthogonal axial aggregation while retaining fine spatial responses through local convolution. Thus, Mamba provides a general sequence-modeling paradigm, whereas AWP serves as a geometry-aware structural operator tailored to the bottleneck stage of the proposed decoder.

C. AWCR Module

At intermediate decoding stages, the deep decoder features contain stronger semantic abstraction, whereas shallow skip features preserve richer spatial details. Direct addition or concatenation of these features therefore often introduces inconsistency between high-level semantics and local structures. To alleviate this issue, we introduce the AWCR block (Algorithm 3) at the ith intermediate decoding stage ($i \in \{2, 3\}$). Given the decoder feature F_D^{i+1} and the FGCR-guided skip feature F_{FGCR}^i, the block produces a refined decoder representation F_D^i.

Specifically, the AWCR first upsamples F_D^{i+1} and projects F_{FGCR}^i into the current decoder feature space using a 3×3 convolution-batch normalization (ConvBN) layer. Two nonnegative normalized weights, w_1 and w_2, are then used to adaptively balance the contributions of the skip feature and the decoder feature, and the weighted sum is refined by a 3×3 ConvBNReLU layer to obtain the initial fused feature F_{WF}^i. Next, F_{WF}^i is processed by two depthwise convolution branches with different receptive fields to capture multiscale spatial context. Their outputs are concatenated and aggregated by average and max operations along the channel dimension, thereby preserving the spatial layout rather than collapsing it into a global descriptor. The resulting two spatial descriptors are fed into a 7×7 convolution followed by a sigmoid function to generate two branch-selection maps, S_1^i and S_2^i, which adaptively select the appropriate receptive field for each spatial region and produce the spatially enhanced context F_{sel}^i. Finally, F_{sel}^i is first mapped by a 1×1 convolution and then used to modulate F_{WF}^i through elementwise interaction, and the resulting feature is further refined by an embedded AWP structure. With this progressive pipeline of cross-layer weighted fusion, multiscale spatial selection, and context refinement, the AWCR block helps reduce semantic inconsistency during decoding while preserving structural continuity and boundary details.

Algorithm 3 Pseudocode of AWCR

Input: The feature-guided context representation $F_{\text{FGCR}}^i \in \mathbb{R}^{B\times C\times H\times W}$, decoder feature $F_D^{i+1} \in \mathbb{R}^{B\times C\times \frac{H}{2}\times \frac{W}{2}}$, $i \in \{2, 3\}$, small constant ε

Output: Refined decoder features $F_D^i \in \mathbb{R}^{B\times C\times H\times W}$

$\widetilde{F}_D^{i+1} \leftarrow Up(F_D^{i+1})$ ▷ Upsample the decoder feature

$\widetilde{F}_{\text{FGCR}}^i \leftarrow ConvBN_{3\times3}(F_{\text{FGCR}}^i)$ ▷ Project F_{FGCR}^i into the decoder channel dimension

*/*Compute non-negative learnable fusion weights*/*

$\mathbf{w} = [w_1, w_2],\ w_j = \dfrac{ReLU(w_j)}{\sum_{l=1}^{2} ReLU(w_l) + \varepsilon},\ j \in \{1, 2\}$

$F_{\text{WF}}^i \leftarrow ConvBNReLU_{3\times3}\left(w_1\widetilde{F}_{\text{FGCR}}^i + w_2\widetilde{F}_D^{i+1}\right)$ ▷ Fuse the two inputs and refine the fused feature

*/*Compute non-negative learnable fusion weights and generate two spatial responses*/*

$\widehat{F}_1^i \leftarrow Conv_{1\times1}\left(DWConv_{5\times5}(F_{\text{WF}}^i)\right)$

$\widehat{F}_2^i \leftarrow Conv_{1\times1}\left(DWConv_{7\times7,d=3}(F_1^i)\right)$

$F_{\text{cat}}^i \leftarrow Cat(\widehat{F}_1^i, \widehat{F}_2^i)$

$F_{\text{avg}}^i \leftarrow Mean(F_{\text{cat}}^i)$ ▷ Concatenate and aggregate

$F_{\text{max}}^i \leftarrow \max(F_{\text{cat}}^i)$

$S^i \leftarrow \sigma\left(Conv_{7\times7}\left(Cat(F_{\text{avg}}^i, F_{\text{max}}^i)\right)\right)$ ▷ Generate branch-wise selection weights

$F^i \leftarrow \widehat{F}_1^i \odot S_1^i + \widehat{F}_2^i \odot S_2^i$ ▷ Axial perception branch

*/*Refine the spatially enhanced feature using the AWP structure*/*

$F_D^i \leftarrow AWP\left(F_{\text{WF}}^i \odot \left(Conv_{1\times1}(F_{\text{sel}}^i)\right)\right)$ ▷ Restore the channel dimension and recalibrate features

Return F_D^i

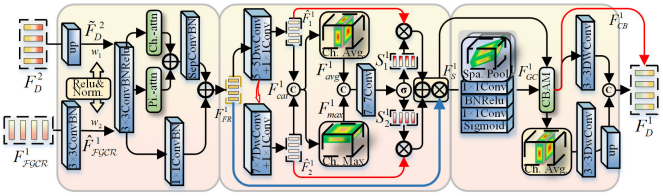


Fig. 4. Overview of the CFR module.

D. CFR Module

In the final decoding stage ($i = 1$), relying on single-layer features is insufficient to balance semantic consistency and local fidelity. Specifically, the upsampled deep feature F_D^2 provides strong semantic discriminability but suffers from severely attenuated boundary details and small-object responses. Conversely, the highest resolution skip connection feature F_{FGCR}^1 retains rich texture and structural information but is highly susceptible to local background noise. To resolve the deep-shallow inconsistency and semantic-detail mismatch, we introduce the CFR module (as shown in Fig. 4) at the final decoding step.

Given inputs F_D^2 and F_{FGCR}^1, we first align them spatially and channelwise. Subsequently, we employ the same adaptive

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weighted fusion strategy and dual-branch spatial selection mechanism as the AWCR module to filter and reorganize local responses across different receptive fields, yielding the selectively enhanced feature F_s^1 (the detailed formulations are omitted as they are identical to AWCR).

To utilize global semantics to constrain local high-resolution responses and mitigate isolated noise, the CFR applies context modulation to F_s^1. A global context descriptor is first extracted via spatial pooling

$$F_{\text{ctx}}^1 = \text{SpatialPool}\left(F_s^1\right). \tag{3}$$

This descriptor is then used for channelwise multiplicative recalibration

$$F_{\text{GC}}^1 = F_s^1 \odot \sigma\big(\text{Conv}_{1\times1}\left(\delta\left(\text{LN}\left(\text{Conv}_{1\times1}\left(F_{\text{ctx}}^1\right)\right)\right)\right)\big) \tag{4}$$

where δ denotes a nonlinear activation and σ is the sigmoid function. Subsequently, a CBAM module performs joint channel and spatial calibration, outputting $F_{\text{CB}}^1 = \text{CBAM}(F_{\text{GC}}^1)$.

To further alleviate scale mismatch during the final recovery phase, F_{CB}^1 is split along the channel dimension into n subfeatures $\{F_{\text{CB}}^{1,k}\}_{k=1}^n$, where $n = 2$ in this work. Each chunk undergoes spatially adaptive modulation

$$F_{\text{sm}}^{1,k} = \begin{cases} \text{DWConv}_{3\times3}\left(F_{\text{CB}}^{1,k}\right), & k = 1 \\ \text{Up}\left(\text{DWConv}_{3\times3}\left(\text{Pool}\left(F_{\text{CB}}^{1,k}\right)\right)\right), & k > 1. \end{cases} \tag{5}$$

Finally, these multiscale features are concatenated and aggregated to produce the refined output

$$F_D^1 = \text{GELU}\big(\text{Conv}_{1\times1}\left(\text{Cat}\left(F_{\text{sm}}^{1,1}, \ldots, F_{\text{sm}}^{1,n}\right)\right)\big) \odot F_{\text{CB}}^1. \tag{6}$$

Unlike AWCR, which focuses on midlevel cross-layer coordination, CFR is tailored for the final high-resolution recovery. By imposing strict global context constraints and local fine-grained calibration on the fused features, the resulting F_D^1 effectively overcomes scale variations and boundary blurring. It is highly optimal as the final input to the segmentation head, particularly benefiting the precise delineation of complex boundaries, small targets, and thin structures.

IV. Data Preparation and Experimental Preparation

A. Materials and Data Preparation

To validate the effectiveness of the proposed FGAR-Net, three challenging high-resolution remote sensing semantic segmentation datasets, namely, ISPRS Vaihingen, ISPRS Potsdam, and WHDLD, were adopted in this study. Fig. 5 presents the representative samples and their annotations from the three datasets. These datasets differ substantially in spatial resolution, scene complexity, and category distribution, thereby providing a comprehensive benchmark for evaluating the proposed method.

The ISPRS Vaihingen dataset is a widely used urban semantic segmentation benchmark. It contains 33 image tiles with a spatial resolution of 9 cm and provides corresponding DSM data. Following the official setting, five foreground categories are considered, including impervious surface, building, low vegetation, tree, and car, together with one clutter category.

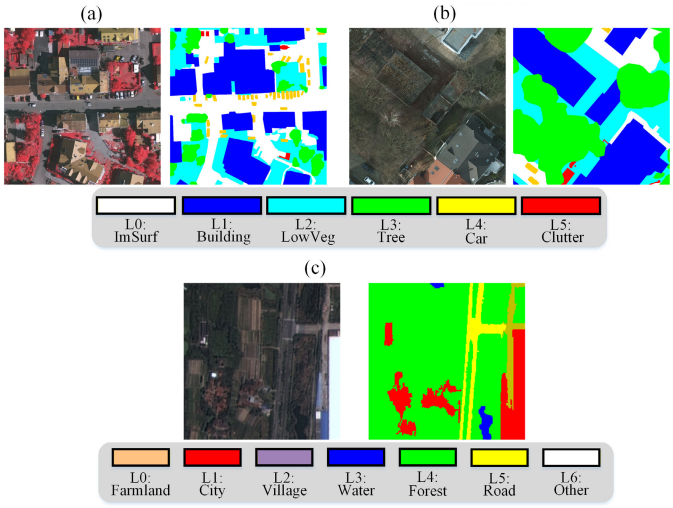


Fig. 5. Dataset samples and attribute labels. (a) ISPRS Vaihingen. (b) ISPR Potsdam. (c) WHDLD.

The original images were first cropped into 1024×1024 patches and then further cropped into 512×512 patches with data augmentation during training.

The ISPRS Potsdam dataset covers a larger urban area with a finer spatial resolution of 5 cm and contains 38 orthophoto tiles of size 6000×6000 pixels, together with DSM data. Its category definitions are consistent with those of Vaihingen. For fair comparison, all images were first divided into 1024×1024 patches and then randomly cropped into 512×512 patches for training.

The WHDLD dataset contains more diverse land-cover scenes and includes categories, such as building, road, pavement, vegetation, bare soil, and water. Compared with the ISPRS datasets, it presents stronger scale variation, more complex background interference, and a more challenging category distribution. In our experiments, the samples were also cropped into 512×512 patches after augmentation for training and evaluation.

B. Implementation Details

All experiments were conducted on an NVIDIA RTX PRO 6000 graphics processing unit (GPU) with 96 GB of memory under Ubuntu 20.04, Python 3.10.19, PyTorch 2.11.0, and compute unified device architecture (CUDA) 12.8. The proposed network was optimized using AdamW with a weight decay of 2.5×10^{-4}. The initial learning rate was set to 6×10^{-4}, while a smaller learning rate of 6×10^{-5} was used for the backbone under a layerwise learning-rate strategy. The learning-rate schedule followed cosine annealing warm restarts with $T_0 = 15$ and $T_{\text{mult}} = 15$. The loss function was defined as an equally weighted combination of soft cross-entropy loss and dice loss. The batch size was set to 8, and all models were trained for 100 epochs. Random flipping and other standard augmentation strategies were used during training, whereas single-scale inference was adopted during testing.

Unless otherwise specified, Swin Transformer was used as the default backbone. ResNet18, ResNet50, and ResNet101 were additionally employed to evaluate the compatibility

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TABLE I

COMPARISON EXPERIMENT RESULTS (%) WITH THE STATE-OF-THE-ART METHODS ON THE ISPRS VAIHINGEN DATASET. THE BOLD VALUES INDICATE THE BEST PERFORMANCE

Method	IoU					mIoU(%)	OA(%)	mF1(%)
	ImSurf	Building	LowVeg	Tree	Car			
FTUnetFormer$_{TGRS2024}$ [30]*	90.26	83.19	70.00	73.56	65.07	76.42	90.38	86.34
ReSMamba$_{TGRS2026}$ [37]*	88.71	95.38	74.29	84.82	83.09	79.10	92.75	85.61
MSLK-EAN$_{ESWA2026}$ [39]*	91.90	94.53	83.05	89.73	87.56	80.92	–	–
IGECNet$_{TGRS2026}$ [29]*	85.72	93.19	66.54	82.11	77.20	80.95	91.33	74.82
MFNet$_{TGRS2025}$ [28]	92.94	89.19	73.64	76.74	77.92	82.09	92.53	89.97
A2FPN$_{TII2022}$ [24]	94.23	92.46	74.37	76.78	80.33	82.14	93.27	90.88
ABCNet$_{ISPRS2020}$ [23]	93.71	92.80	74.83	76.85	72.51	82.14	93.27	89.92
BANet$_{RS2021}$ [58]	92.73	89.67	73.01	76.32	80.44	82.43	92.46	90.18
FreDNet$_{TGRS2026}$ [53]*	–	–	–	–	–	83.31	89.57	91.33
FSSFFormer$_{IF2026}$ [42]*	–	–	–	–	–	83.45	91.21	92.12
UNetFormer$_{ISPRS2022}$ [55]*	93.85	92.76	74.37	77.04	80.08	83.62	93.30	90.87
DCSwin$_{GRSL2024}$ [56]*	94.39	93.94	74.82	76.72	81.25	84.22	93.60	91.21
FGAR-Net (Swin)	94.62	93.64	75.79	78.01	83.77	**85.17**	**93.85**	**91.80**

TABLE II

COMPARISON EXPERIMENTAL RESULTS (%) WITH THE STATE-OF-THE-ART METHODS ON THE ISPRS POTSDAM DATASET. THE BOLD MEANS THE BEST PERFORMANCE

Method	IoU					mIoU(%)	OA(%)	mF1(%)
	ImSurf	Building	LowVeg	Tree	Car			
ReSMamba$_{TGRS2026}$ [37]*	88.71	95.38	74.29	84.82	83.09	79.10	**92.75**	85.61
IGECNet$_{TGRS2026}$ [29]*	79.13	90.74	68.21	72.53	87.45	79.71	86.53	81.43
FTUnetFormer$_{TGRS2024}$ [30]#	84.57	88.39	72.84	71.85	87.17	80.96	88.17	89.31
FreDNet$_{TGRS2026}$ [53]*	–	–	–	–	–	82.76	90.53	92.07
FSSFFormer$_{IF2026}$ [42]*	–	–	–	–	–	83.86	91.92	**93.37**
ABCNet$_{ISPRS2021}$ [23]*	86.39	91.33	77.69	79.41	89.36	84.84	90.59	91.70
MFNet$_{TGRS2025}$ [28]*	87.13	92.31	76.95	79.56	91.72	85.53	90.75	92.07
UNetFormer$_{ISPRS2022}$ [55]*	88.33	93.15	77.35	79.26	92.52	86.12	91.17	92.40
A2FPN$_{TII2022}$ [24]#	88.37	92.81	77.44	79.79	92.52	86.19	91.22	92.45
BANet$_{RS2021}$ [58]*	88.64	93.32	78.10	80.58	91.87	86.51	91.52	92.65
DCSwin$_{GRSL2024}$ [56]*	89.52	94.20	78.69	80.98	93.14	87.31	91.89	93.10
FGAR-Net (Swin)	89.39	94.21	78.84	81.04	94.27	**87.55**	91.89	93.23

of the proposed framework with different encoders. For all reimplemented methods marked with "#," the experiments on the three datasets were conducted using the same hardware and software environment, dataset partitions, image preprocessing, data augmentation, training schedule, and evaluation code. The results marked with "*" were directly quoted from the corresponding publications because their official implementations were not publicly available at the time of our experiments. These literature-reported results are included as reference comparisons and should not be interpreted as strictly controlled comparisons with the reimplemented methods. For all reproduced experiments, the train/validation/test partitions followed the protocols adopted by UNetFormer [58], DCSwin [59], and CADANet [60].

C. Evaluation Metrics

To quantitatively evaluate the performance of the proposed method for high-resolution remote sensing image semantic segmentation, three widely used metrics are adopted, including overall accuracy (OA), mean $F1$-score (mF1), and mean intersection over union (mIoU). Among them, the OA measures the overall pixelwise classification accuracy over all classes, while mF1 and mIoU are mainly used to assess the segmentation quality of foreground classes

$$
\begin{aligned}
OA &= \frac{\sum_{c=1}^{C} TP_c}{\sum_{c=1}^{C} (TP_c + FN_c)} \\
mF1 &= \frac{1}{C} \sum_{c=1}^{C} \frac{2P_c R_c}{P_c + R_c} \\
mIoU &= \frac{1}{C} \sum_{c=1}^{C} \frac{TP_c}{TP_c + FP_c + FN_c}
\end{aligned} \tag{7}
$$

where P_c and R_c denote the precision and recall of class c, respectively, and TP_c, FP_c, and FN_c represent the numbers of true positive, false positive, and false negative pixels for class c, respectively.

V. EXPERIMENTAL RESULTS AND ANALYSIS

A. Contrast Evaluation for the State-of-the-Art Methods

On the ISPRS Vaihingen dataset (Table I), FGAR-Net (Swin) achieves 85.17% mIoU, 93.85% OA, and 91.80% mF1, outperforming strong competitors such as DCSwin, UNetFormer, and FSSFFormer. A closer look at the classwise results shows that the proposed method attains 83.77% IoU on the car category, which is clearly higher than 80.08% from UNetFormer and 81.25% from DCSwin. Meanwhile, it also remains highly competitive on categories, such as impervious surface, building, and tree. These results suggest that FGAR-Net is beneficial not only for recovering small objects and local structures but also for maintaining semantic integrity over dominant land-cover regions. This behavior is consistent with the decoding design of the proposed framework. Specifically, the FGCR introduces frequency priors into the skip pathways and selectively recalibrates shallow features before transmission, thereby suppressing redundant background responses. The AWP module jointly models regional semantics, long-range structural dependencies, and local details at the bottleneck stage, while the CFR module further constrains local responses at the final high-resolution stage, benefiting boundary restoration and fine-grained detail recovery.

On the ISPRS Potsdam dataset (Table II), FGAR-Net achieves 87.55% mIoU and 93.23% mF1, outperforming DCSwin by 0.24 and 0.13 percentage points, respectively, while maintaining the same OA of 91.89%. The gain is mainly concentrated in categories with relatively small spatial coverage or fine structural characteristics. In particular, the IoU of car increases from 93.14% to 94.27%, while low vegetation and tree improve by 0.15 and 0.06 percentage points, respectively. Since mIoU assigns equal weight to each class, the 1.13-point improvement for car alone contributes approximately 0.23 percentage points to the overall mIoU. By contrast, OA is dominated by pixel-abundant classes, such as impervious surfaces and buildings, for which the two methods yield nearly identical performance; consequently, the improvements on small objects have only a marginal effect on the overall pixel accuracy. From the model perspective, FGCR suppresses redundant shallow-frequency responses before feature transmission, AWCR progressively coordinates semantic and spatial information across decoding stages, and CFR further refines local responses at the final high-resolution stage. Their complementary effects primarily enhance small-object discrimination and fine class-transition recovery, thereby improving class-balanced metrics without substantially altering OA.

On the WHDLD dataset (Table III), FGAR-Net again achieves the best performance, with 65.75% mIoU and 78.01%

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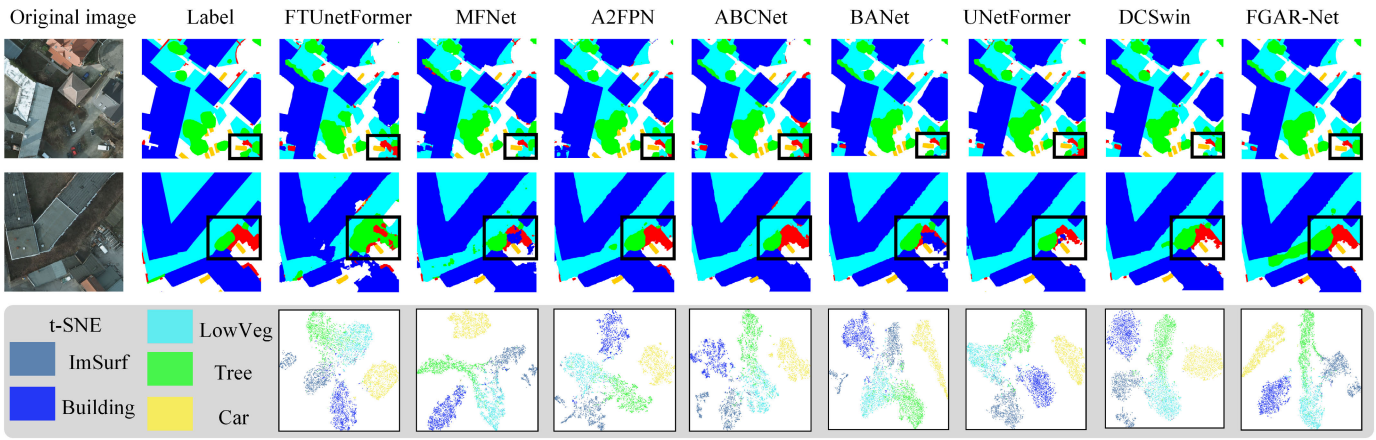


Fig. 6. Visual comparison of the experimental results on the ISPRS Vaihingen dataset.

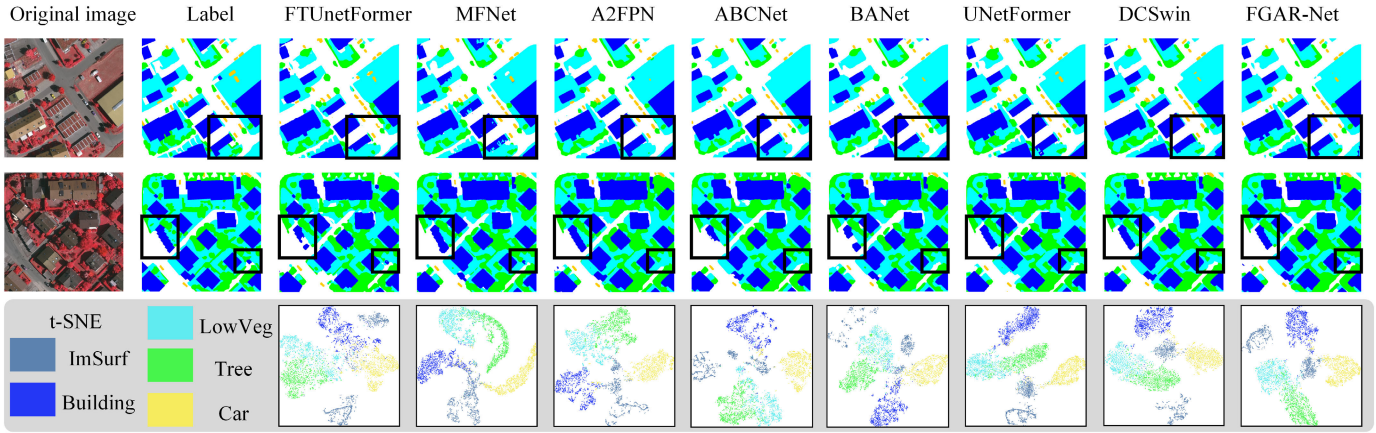


Fig. 7. Visual comparison of the experimental results on the ISPRS Potsdam dataset.

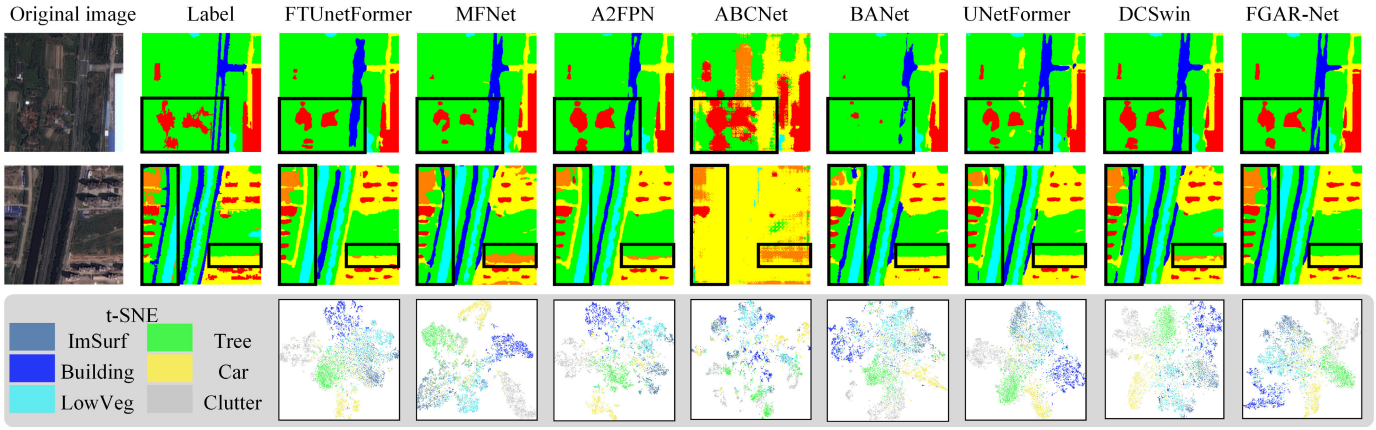


Fig. 8. Visual comparison of the experimental results on the WHDLD dataset.

mF1, outperforming DCSwin, CADANet, and MF-Mamba. More specifically, the gains on Building and Road are more evident, reaching 60.39% and 65.29%, respectively, which suggests that the proposed method remains effective in scenes with severe scale variation and stronger demands on structural continuity. From the perspective of model design, this result is consistent with the ability of AWP to model long-range continuous structures at the bottleneck stage, as well as the role of AWCR in progressively coordinating cross-layer semantics during decoding. Compared with strategies that rely mainly on local convolution enhancement or single-form attention modeling, FGAR-Net maintains more stable structural representation under the coexistence of multiscale objects and cluttered backgrounds.

The visual comparisons in Figs. 6–8 further support the above observations. Compared with the competing methods,

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TABLE III

COMPARISON EXPERIMENTAL RESULTS (%) WITH THE STATE-OF-THE-ART METHODS ON THE WHDLD DATASET. THE BOLD MEANS THE BEST PERFORMANCE

Method	IoU(%)						mIoU(%)	OA(%)	mF1(%)
	Building	Road	Pavement	Vegetation	Bare soil	Water			
ABCNet$_{ISPRS2021}$ [23]*	34.20	0.10	24.29	51.47	24.41	84.32	36.47	62.76	48.17
FTUnetFormer$_{TGRS2024}$ [30]#	55.03	53.94	41.40	78.89	41.18	93.06	60.59	82.27	73.76
HMA-UNet$_{AA2026}$ [27]*	–	–	–	–	–	–	60.71	72.40	73.90
A2FPN$_{IJRS2022}$ [24]#	54.04	58.18	42.87	78.37	41.75	92.10	61.22	82.39	74.40
BANet$_{RS2021}$ [58]*	55.09	57.23	42.36	79.08	40.39	93.49	61.27	82.80	74.31
UNetFormer$_{ISPRS2022}$ [55]*	56.87	60.25	43.22	79.60	41.87	93.39	62.53	83.45	75.39
MFNet$_{TGRS2025}$ [28]*	56.59	60.36	44.42	79.85	44.11	93.74	63.18	83.51	75.98
MF-Mamba$_{TGRS2025}$ [36]*	–	–	–	–	–	–	64.79	**85.26**	77.07
CADANet$_{TGRS2025}$ [57]*	57.97	62.86	45.31	83.55	44.02	94.93	64.77	85.24	77.55
DCSwin$_{GRSL2024}$ [56]*	58.53	64.73	45.91	80.17	45.90	93.96	64.87	84.31	77.36
FGAR-Net (Swin)	60.39	65.29	47.11	81.20	45.75	94.79	**65.75**	85.05	**78.01**

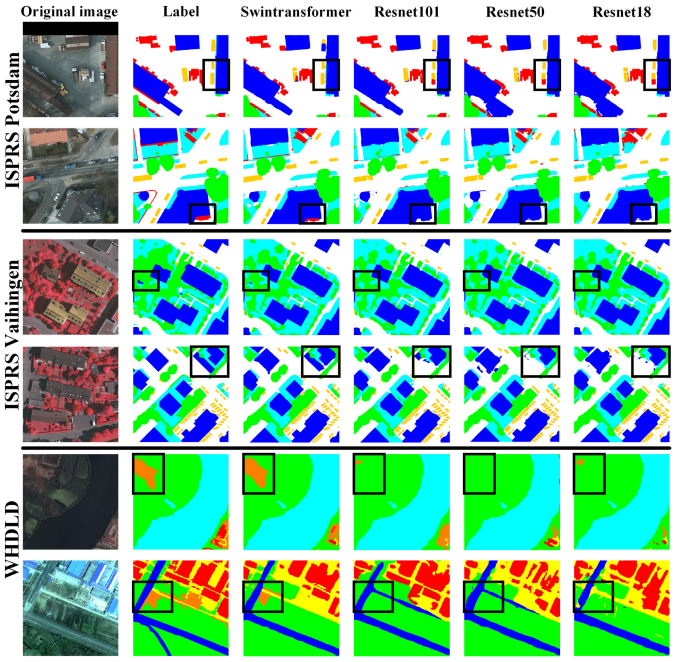


Fig. 9. Visual comparison of the experimental results using different backbones on the Vaihingen dataset.

FGAR-Net generally produces more complete building contours, more continuous road or impervious-surface structures, and clearer class-transition boundaries in complicated regions. It also exhibits fewer missed predictions and false responses in locally crowded areas and between adjacent confusing categories. These observations indicate that the advantage of FGAR-Net is not limited to improvements in overall metrics but is also reflected in the joint enhancement of structural continuity, boundary-detail recovery, and robustness to cluttered backgrounds. In light of the proposed design, such behavior is consistent with the selective feature delivery of FGCR, the structural modeling of AWP, the stagewise semantic coordination of AWCR, and the final high-resolution refinement of CFR.

To verify whether the proposed framework depends on a specific encoder, we further evaluated it with four backbones, including ResNet18, ResNet50, ResNet101, and Swin Transformer (Table IV and Fig. 9). The results show that FGAR-Net maintains stable performance across different backbones and generally benefits from stronger feature encoders, suggesting that it provides a relatively independent

TABLE IV

COMPUTATIONAL COMPLEXITY ANALYSIS MEASURED ON RTX PRO 6000 GPU (96 GB). THE SEGMENTATION RESULTS ARE REPORTED ON THE WHDLD, ISPRS VAIHINGEN, AND ISPRS POTSDAM DATASETS

Backbone	Parms	GFLOPs	FPS	WHDLD			ISPRS Vaihingen			ISPRS Potsdam		
				mIoU	OA	mF1	mIoU	OA	mF1	mIoU	OA	mF1
ResNet18	20.04	50.11	77.37	61.87	83.15	74.88	86.40	91.46	92.59	86.40	91.46	92.44
ResNet50	36.52	74.87	64.32	62.54	83.64	75.32	87.49	91.98	94.05	87.49	91.98	93.20
ResNet101	55.51	94.32	53.34	62.95	83.49	75.81	87.06	91.74	92.96	87.06	91.74	92.96
Swin Transformer	96.90	131.93	37.77	**65.75**	**85.05**	**78.01**	**85.17**	**93.85**	**91.80**	**87.55**	**91.89**	**93.23**

TABLE V

COMPARISON OF SEGMENTATION PERFORMANCE, MODEL COMPLEXITY, AND INFERENCE EFFICIENCY OF DIFFERENT METHODS ON THE ISPRS VAIHINGEN DATASET

Method	Parms(M)	GFLOPs	FPS	Latency	Weight Memory	mIoU(%)	OA(%)	mF1(%)
FTUnetFormer$_{TGRS2024}$ [30]#	36.17	64.71	82.25	12.16	137.99	76.42	90.38	86.34
IGECNet$_{TGRS2026}$ [29]*	1916.52	316.66	–	–	–	80.95	91.33	74.82
MFNet$_{TGRS2025}$ [28]*	35.86	54.20	132.74	7.53	136.80	82.09	92.53	89.97
A2FPN$_{IJRS2022}$ [24]#	12.16	13.53	324.32	3.08	46.39	82.14	93.27	90.88
ABCNet$_{ISPRS2021}$ [23]*	13.68	15.67	177.01	5.64	52.20	82.14	93.27	89.92
BANet$_{RS2021}$ [58]*	12.73	15.15	94.06	10.63	48.56	82.43	92.46	90.18
FreDNet$_{TGRS2026}$ [53]*	39.84	181.60	–	24.80	–	83.31	89.57	91.33
FSSFFormer$_{IF2026}$ [42]*	58.50	48.27	–	20.10	–	83.45	91.21	92.12
UNetFormer$_{ISPRS2022}$ [55]*	11.72	11.76	181.87	5.46	44.73	83.62	93.30	90.87
DCSwin$_{GRSL2024}$ [56]*	66.95	72.18	54.99	18.19	255.39	84.22	93.60	91.21
FGAR-Net (ResNet18)	20.04	50.11	77.37	12.93	76.46	84.11	93.51	91.16
FGAR-Net (ResNet50)	36.52	74.87	64.32	15.55	139.31	84.56	93.60	91.43
FGAR-Net (ResNet101)	55.51	94.32	53.34	18.75	211.75	85.04	93.84	91.72
FGAR-Net (Swin)	96.90	131.93	37.77	26.47	369.67	**85.17**	**93.85**	**91.80**

recovery-and-refinement mechanism rather than relying merely on a single strong encoder. Meanwhile, the complexity analysis shows that FGAR-Net offers multiple practical accuracy-cost tradeoffs under different backbone settings. The ResNet-based variants already achieve competitive results with acceptable parameter size and inference speed, whereas the Swin-based version further improves structural recovery in complicated scenes. In particular, on the ISPRS Vaihingen efficiency comparison, FGAR-Net (ResNet18) achieves 84.11% mIoU with only 20.04M parameters and 77.37 frames/s, already surpassing several existing methods. Although FGAR-Net (Swin) introduces higher parameter and computational costs, it delivers the best overall performance in this section.

To further evaluate the tradeoff among segmentation accuracy, model complexity, and inference efficiency, we compare different methods on the ISPRS Vaihingen dataset in Table V. The FGAR-Net (Swin) achieves the best overall performance, with 85.17% mIoU and 93.85% OA. Although this variant introduces higher computational cost, with 96.90M parameters and 131.93 GFLOPs, its complexity remains substantially lower than that of extremely large models, such as IGECNet (1916.52M parameters and 316.66 GFLOPs), while still delivering higher mIoU. This suggests that the performance gain of the proposed framework is not obtained by relying on excessive model scale.

In addition, FGAR-Net shows good adaptability across different backbone capacities. As the backbone changes from ResNet18 to Swin Transformer, the mIoU increases steadily from 84.11% to 85.17%. Notably, the lightweight FGAR-Net (ResNet18) already achieves 84.11% mIoU with only 20.04M parameters, 50.11 GFLOPs, and 77.37 frames/s, outperforming MFNet, FreDNet, and UNetFormer in mIoU. These results indicate that the proposed asymmetric progressive decoding framework provides a favorable balance between accuracy and

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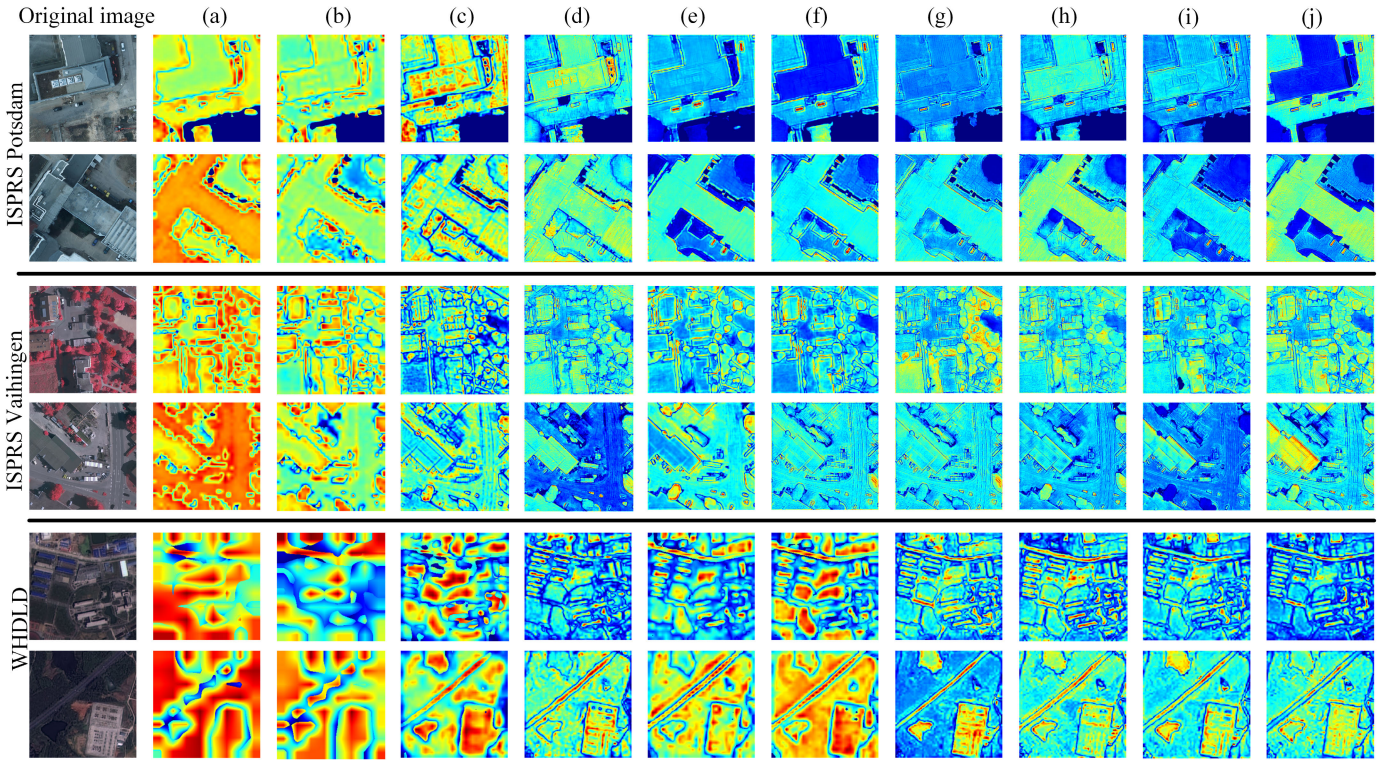


Fig. 10. Visual comparison of the ablation results under different module configurations on the three datasets. (a)–(j) Represent ten random cases from each of the three datasets.

TABLE VI

ABLATION RESULTS UNDER DIFFERENT MODULE CONFIGURATIONS ON THREE DATASETS. THE BOLD VALUES INDICATE THE BEST PERFORMANCE

ID	FGCR	CFR	AWCR	B2 AWCR B3	AWP	WHDLD			Vaihingen			Potsdam			Complexity		
						mIoU	OA	mF1	mIoU	OA	mF1	mIoU	OA	mF1	Param	GF	FPS
(a)	✗	✗	✗	✗	✗	55.18	78.06	69.38	80.37	92.80	88.72	84.72	91.30	91.63	87.60	87.26	67.30
(b)	✓	✗	✗	✗	✓	55.52	78.11	69.71	80.38	92.72	88.75	85.30	91.42	91.97	89.24	87.65	47.15
(c)	✓	✓	✓	✓	✗	60.08	81.74	73.40	83.90	93.65	91.01	85.37	90.72	91.98	95.36	104.83	42.40
(d)	✗	✓	✗	✗	✗	61.96	83.06	74.89	84.40	93.71	91.34	85.87	91.04	92.27	92.10	123.29	56.43
(e)	✗	✗	✓	✓	✗	64.39	84.18	76.96	84.63	93.69	91.46	87.40	91.88	93.16	93.72	104.44	48.73
(f)	✓	✗	✓	✓	✗	64.50	84.12	77.03	84.48	93.60	91.37	87.37	91.81	93.14	93.89	104.45	40.31
(g)	✓	✓	✗	✗	✓	65.37	84.89	77.70	85.03	93.79	91.70	87.34	91.79	93.12	95.44	131.55	40.03
(h)	✓	✓	✓	✗	✓	65.73	85.16	77.98	84.84	93.77	91.59	87.40	91.78	93.15	92.27	123.30	42.14
(i)	✓	✓	✗	✓	✓	65.74	85.14	78.00	84.67	93.72	91.50	87.38	91.83	93.14	93.74	123.68	41.22
(j)	✓	✓	✓	✓	✓	**65.75**	**85.05**	**78.01**	**85.17**	**93.85**	**91.80**	**87.55**	**91.89**	**93.23**	96.90	131.93	37.77

efficiency and does not rely solely on enlarging the backbone capacity.

B. Ablation Study

To verify the individual contributions of the components in FGAR-Net and their collaborative effect, we first conducted ablation experiments under different module configurations on the three datasets. As shown in Table VI, the baseline configuration (a) achieves 55.18%, 80.37%, and 84.72% mIoU on WHDLD, ISPRS Vaihingen, and ISPRS Potsdam, respectively, while the full model (j) further improves them to 65.75%, 85.17%, and 87.55%, corresponding to the gains of 10.57, 4.80, and 2.83 points. Meanwhile, the model complexity also increases as more components are introduced: the parameter size rises from 87.60M to 96.90M, the GFLOPs increase from 87.26 to 131.93, and the FPS decreases from 67.30 to 37.77. This indicates that the performance gain of FGAR-Net is not obtained for free but is achieved by introducing a more complete recovery pipeline spanning selective feature delivery FGCR, bottleneck structural perception AWP, cross-level coordination AWCR, and final high-resolution refinement CFR. A closer comparison of representative intermediate configurations reveals this accuracy–efficiency evolution more clearly. For example, only AWCR (e), with 93.72M parameters and 104.44 GFLOPs, already improves the mIoU to 64.39%, 84.63%, and 87.40% on the three datasets. The No-AWP (g), with 95.44M parameters and 131.55 GFLOPs, further raises the mIoU to 65.37% on WHDLD and 85.03% on Vaihingen. Compared with the No-AWP configuration (g), the full model (j) introduces only 1.46M additional parameters and 0.38 GFLOPs, while further improving the mIoU on all three datasets. These consistent gains confirm that AWP provides complementary structural modeling at the bottleneck stage by jointly capturing regional semantics, long-range axial dependencies, and local details. Together with the visual comparisons in Fig. 10, this additional cost mainly translates into more stable boundary recovery, more continuous thin structures, and fewer false predictions in cluttered backgrounds and is therefore reasonable overall.

To directly verify the rationality of the module–stage assignment, we compared the proposed decoder with three controlled variants, including the uniform deployment of AWCR, the uniform deployment of CFR, and the exchange of AWCR and CFR between the intermediate and final decoding stages. As shown in Table VII, applying either AWCR or CFR to all decoding stages results in lower performance on most segmentation metrics, indicating that a homogeneous

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TABLE VII

ABLATION RESULTS OF DIFFERENT MODULE–STAGE ASSIGNMENTS FOR THE STAGE-AWARE DECODER ON THREE DATASETS. DECODERS 1 AND 4 DENOTE THE FINAL HIGH-RESOLUTION AND BOTTLENECK STAGES, RESPECTIVELY. THE BOLD VALUES INDICATE THE BEST PERFORMANCE

Decoder 1	Decoder 2	Decoder 3	Decoder 4	WHDLD			ISPRS Vaihingen			ISPRS Potsdam			Params	GFLOPs	FPS
				mIoU	OA	mF1	mIoU	OA	mF1	mIoU	OA	mF1			
AWCR	*AWCR*	*AWCR*	*AWCR*	65.62	85.06	77.92	84.73	93.84	91.53	87.55	91.82	93.23	97.95	153.07	38.96
CFR	*CFR*	*CFR*	*CFR*	65.44	85.05	77.75	84.74	93.59	91.53	87.20	91.77	93.05	94.11	125.00	38.41
AWCR	*CFR*	*CFR*	*AWP*	65.73	**85.17**	77.99	84.72	93.72	91.51	87.40	**91.95**	93.16	95.98	146.44	39.77
CFR	*AWCR*	*AWCR*	*AWP*	**65.75**	85.05	**78.01**	**85.17**	**93.85**	**91.80**	**87.55**	91.89	**93.23**	96.90	131.93	37.77

TABLE VIII

ABLATION RESULTS OF REPLACING DIFFERENT CHANNEL SELECTION METHODS

Method	Parms	GFLOPs	FPS	WHDLD			ISPRS Vaihingen			ISPRS Potsdam		
				mIoU	OA	mF1	mIoU	OA	mF1	mIoU	OA	mF1
SE [62]	97.60	131.93	49.57	64.70	84.34	77.17	84.48	93.65	91.37	87.52	91.96	93.22
CBAM [63]	97.08	131.93	48.13	65.17	84.61	77.54	84.62	93.68	91.46	87.39	91.80	93.15
SCSA [64]	96.75	131.92	45.80	64.22	83.92	76.79	84.71	93.62	91.52	87.50	91.86	93.33
FGCR	96.90	131.93	37.77	**65.75**	**85.05**	**78.01**	**85.17**	**93.85**	**91.80**	**87.55**	**91.89**	**93.23**

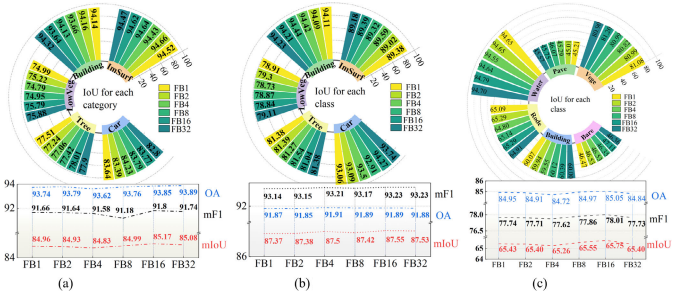


Fig. 11. Performance comparison of different frequency branches of module on three datasets. (a) ISPRS Vaihingen. (b) ISPR Potsdam. (c) WHDLD.

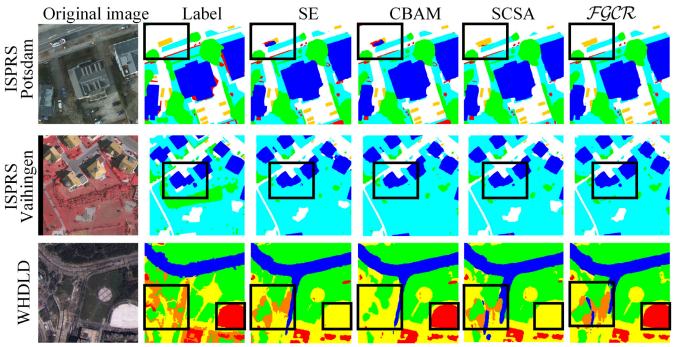


Fig. 12. Visualization results of the ablation study with different channel selection method replacements.

refinement mechanism cannot simultaneously accommodate the heterogeneous recovery requirements at different feature resolutions. Exchanging the positions of AWCR and CFR also degrades the overall performance, with the mIoU on Vaihingen decreasing from 85.17% to 84.72%. In contrast, the proposed configuration, AWP at the bottleneck, AWCR at the two intermediate stages, and CFR at the final high-resolution stage, achieves the highest mIoU on WHDLD and Vaihingen and ties for the best result on Potsdam. It also yields the highest mF1 on WHDLD and Vaihingen while matching the best mF1 on Potsdam. These results demonstrate that AWCR is better suited to progressive cross-level semantic coordination at intermediate resolutions, whereas CFR is more effective for boundary- and detail-oriented refinement at the final high-resolution stage, thereby providing direct evidence for the necessity of the proposed stage-aware decoding strategy.

We investigated the effect of the frequency-branch (FB) setting in FGCR, considering both performance tendency and complexity implication. Fig. 11 and the corresponding results show that the branch design is performance-sensitive, but the gain does not increase monotonically with more branches. On Vaihingen, the model reaches its best result at a moderate branch setting, with FB16 achieving 85.17% mIoU and 91.80% mF1; when the number is further increased to FB32, the mIoU drops to 85.08%, suggesting that excessive branches do not continue to improve effective representation. A similar tendency can also be observed on Potsdam and WHDLD, where the best results are also obtained within a moderate branch range, while both lower and higher settings cause slight degradation. This suggests that too few frequency branches limit the richness of frequency-aware representation and weaken the discrimination between shallow redundancy and high-frequency structural details, whereas too many branches may introduce more dispersed responses and lead to saturated gains.

To further verify the necessity of the current FGCR design, we replaced it with three commonly used channel selection methods (as shown in Fig. 12 and Table VIII), namely, SE [61], CBAM [62], and SCSA [63], and jointly compared segmentation performance and complexity.

As shown in Fig. 13, the different kernel settings have only marginal differences in parameter size, remaining within 96.88M–96.92M, but they still exhibit an observable trade-off among GFLOPs, FPS, and segmentation performance. Among them, the $5 \times 5 + 7 \times 7$ combination achieves the best overall results, reaching 65.75%, 85.17%, and 87.55% mIoU on WHDLD, Vaihingen, and Potsdam, respectively, with corresponding mF1 values of 78.01%, 91.80%, and 93.23%. Its GFLOPs and FPS are 131.93 and 37.77. By comparison, the smaller $3\times3+3\times3$ setting has slightly lower computation (131.85 GFLOPs), but only 35.17 frames/s, and its mIoU drops to 84.90% and 87.41% on Vaihingen and Potsdam. The larger $7 \times 7 + 7 \times 7$ setting slightly improves FPS to

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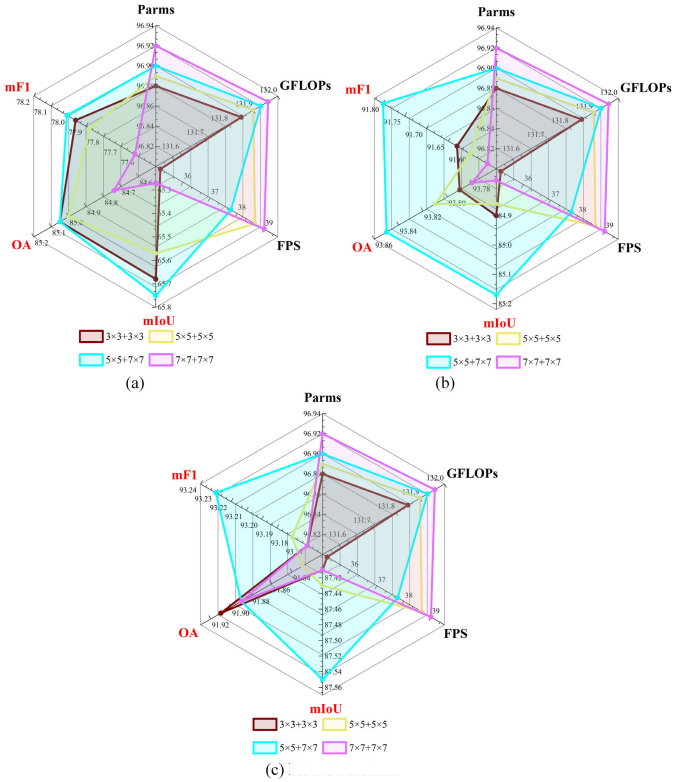


Fig. 13. Visualization results of different size convolution kernels of AWCR module on three datasets. (a) WHDLD. (b) ISPRS Vaihingen. (c) ISPR Potsdam.

39.00 but increases GFLOPs to 131.96 and lowers the mIoU on all three datasets to 65.27%, 84.78%, and 87.41%. The $5 \times 5 + 5 \times 5$ setting lies between these two extremes but still remains below $5 \times 5 + 7 \times 7$ overall. These results indicate that the adopted kernel combination is not simply a larger receptive-field choice, but a more balanced design between local structure preservation and contextual coordination range. The visual results in Fig. 13 also support this interpretation: smaller kernels tend to show insufficient semantic coordination over large regions, whereas overly large kernels are more likely to blur boundaries and weaken small-object responses.

VI. Conclusion

This article investigates the challenges of high-resolution remote sensing image semantic segmentation, particularly focusing on fine-grained category confusion, the coexistence of multiscale objects, and the degradation of boundary details. To address the dilemma that conventional feature fusion strategies struggle to balance deep global semantics with shallow local details, we propose FGAR-Net. Moving beyond the conventional paradigm of simply stacking isolated modules, this method constructs an asymmetric stepwise decoding pipeline that integrates selective feature delivery, bottleneck structural modeling, and stage-aware recovery. Specifically, the framework introduces DCT priors into skip pathways to selectively filter background redundancy. At the bottleneck stage, it anchors regional semantics, long-range structural dependencies, and local details jointly. During the decoding process, it implements cross-layer semantic coordination and context-constrained high-resolution local reconstruction tailored to different feature depths. Extensive experiments on the ISPRS Vaihingen, ISPRS Potsdam, and WHDLD datasets demonstrate that FGAR-Net exhibits high stability in complex scene parsing. It not only effectively alleviates cross-layer semantic mismatch but also shows distinct advantages in identifying small targets and preserving complex boundaries, with ablation studies further confirming the necessity and synergy of the stage-aware design.

The findings of this study indicate that improving fine-grained semantic parsing in high-resolution remote sensing scenes relies not only on upgrading feature extraction backbones but also on explicitly organizing and assigning stage-specific tasks within the decoding recovery process. This approach provides an effective paradigm for decoder design in dense prediction tasks for remote sensing imagery. Despite the superior performance achieved by the proposed framework, certain limitations remain. First, pursuing optimal local detail recovery inevitably introduces additional computational overhead in the stage-aware decoding mechanism, making it less amenable to lightweight deployment on edge devices. Second, the generalization capability of the model across different resolutions or sensor domains requires broader validation. Future research will primarily focus on two directions: exploring more efficient and lightweight stage-aware decoding operators to reduce computational costs while maintaining fine-grained recovery accuracy, and incorporating cross-domain adaptation mechanisms or self-supervised learning strategies to further enhance the robustness of this decoding framework in multi-source heterogeneous remote sensing data and more complex open scenarios.

References

[1] Y. Li et al., "MFVNet: A deep adaptive fusion network with multiple field-of-views for remote sensing image semantic segmentation," *Sci. China Inf. Sci.*, vol. 66, no. 4, Apr. 2023, Art. no. 140305.

[2] Y. Zhou, A. Ma, J. Wang, Z. Chen, and Y. Zhong, "Remote sensing meta modal representation for missing modality land cover mapping: From EarthMiss dataset to MetaRS method," *Remote Sens. Environ.*, vol. 333, Jan. 2026, Art. no. 115132.

[3] C. Wen et al., "Remote sensing retrieval-augmented generation: Bridging remote sensing imagery and comprehensive knowledge with a multi-modal dataset and retrieval-augmented generation model," *IEEE Geosci. Remote Sens. Mag.*, vol. 14, no. 2, pp. 85–103, Apr. 2026.

[4] W. Huang, H. Li, S. Zhang, and F. Deng, "Reducing semantic ambiguity in open-vocabulary remote sensing image segmentation via knowledge graph-enhanced class representations," *ISPRS J. Photogramm. Remote Sens.*, vol. 231, pp. 837–853, Jan. 2026.

[5] J. Xie et al., "Global vision-language feature interaction enhanced by object-context association for remote sensing visual grounding," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5641815.

[6] Z. Li, J. Hu, B. Yang, K. Wu, and J. Wu, "MNDAF-Net: Multiscale neighborhood decoupling and attribute-focused fusion for dual-temporal remote sensing image scene change detection network," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5640916.

[7] Z. Li, J. Hu, K. Wu, J. Miao, and J. Wu, "Comprehensive attribute difference attention network for remote sensing image semantic understanding," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5602716.

[8] Z. Li, J. Hu, K. Wu, J. Miao, and J. Wu, "Adjacent-atrous mechanism for expanding global receptive fields: An end-to-end network for multi-attribute scene analysis in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5523119.

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[9] P. Guo et al., "SCIIENet: Shared and complementary information interaction enhancement network for self-supervised multimodal remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5512617.

[10] J. Li, K. Zheng, Z. Li, L. Gao, and X. Jia, "X-shaped interactive autoencoders with cross-modality mutual learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518317.

[11] P. Liu et al., "CGCNet: Road extraction from remote sensing image with compact global context-aware," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5638312.

[12] J. Liu, M. Cao, and S. Chen, "A gradual coarse-to-fine framework for irregularly sampled multivariate time series analysis," *Sci. China Inf. Sci.*, vol. 69, no. 3, Mar. 2026, Art. no. 132105.

[13] Z. Li et al., "FTransMamba: A multi-stage fusion transformer and mamba modeling for multimodal remote sensing scene understanding," *Pattern Recognit.*, vol. 179, Nov. 2026, Art. no. 113625.

[14] H. Lv, X. Yan, H. Zhang, C. Shi, and R. Wang, "MLGFENet: Multiscale local–global feature enhancement network for high-resolution remote sensing image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 17019–17036, 2025.

[15] Z. Li et al., "HDC-Net: A multimodal remote sensing semantic segmentation network with hierarchical dual-stream fusion and cross-token interaction," *Knowl.-Based Syst.*, vol. 337, Mar. 2026, Art. no. 115416.

[16] Y. Gao et al., "PyS^{2}CF-Mamba: A pyramid spatial–spectral competitive fusion mamba network for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 23, 2026, Art. no. 5507805.

[17] G. Gao, G. Yang, Y. Wang, L. Yao, X. Zhang, and G. Li, "Oriented decoupling target detection method for SAR image based on multi-channel localization and soft thresholding," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 36, no. 1, pp. 219–232, Jan. 2026.

[18] J. Li, K. Zheng, L. Gao, L. Ni, M. Huang, and J. Chanussot, "Model-informed multistage unsupervised network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 132105.

[19] Z. Yang, H. Yao, Q. Li, W. Ni, J. Wu, and Q. Wang, "Semantic-spatial feature refinement network for road extraction from remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5609710.

[20] J. Li, K. Zheng, L. Gao, Z. Han, Z. Li, and J. Chanussot, "Enhanced deep image prior for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504218.

[21] H. Zhou et al., "Exploring self-image and cross-image consistency learning for remote sensing burned area segmentation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 36, no. 3, pp. 2924–2937, Mar. 2026.

[22] Y. Qu, J. Li, and X. Huang, "AnyTime-CD: Self-supervised change detection in arbitrary-length dense time series of high-resolution remote sensing images," *Remote Sens. Environ.*, vol. 334, Mar. 2026, Art. no. 115239.

[23] R. Li, S. Zheng, C. Zhang, C. Duan, L. Wang, and P. M. Atkinson, "ABCNet: Attentive bilateral contextual network for efficient semantic segmentation of fine-resolution remotely sensed imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 181, pp. 84–98, Nov. 2021.

[24] R. Li, L. Wang, C. Zhang, C. Duan, and S. Zheng, "A^2-FPN for semantic segmentation of fine-resolution remotely sensed images," *Int. J. Remote Sens.*, vol. 43, no. 3, pp. 1131–1155, Feb. 2022.

[25] L. Fang, P. Zhou, X. Liu, P. Ghamisi, and S. Chen, "Context enhancing representation for semantic segmentation in remote sensing images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 3, pp. 4138–4152, Mar. 2024.

[26] Y. Gao, X. Luo, X. Gao, W. Yan, X. Pan, and X. Fu, "Semantic segmentation of remote sensing images based on multiscale features and global information modeling," *Expert Syst. Appl.*, vol. 249, Sep. 2024, Art. no. 123616.

[27] L. Chen et al., "Research on remote sensing image object segmentation using a hybrid multi-attention mechanism," *Appl. Sci.*, vol. 16, no. 2, p. 695, Jan. 2026.

[28] X. Ma, X. Zhang, M.-O. Pun, and B. Huang, "A unified framework with multimodal fine-tuning for remote sensing semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5405015.

[29] H. Zhang et al., "IGECNet: An interaction-guided multimodal fusion network with elevation constraint for remote sensing semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5604621.

[30] X. Ma, Q. Wu, X. Zhao, X. Zhang, M.-O. Pun, and B. Huang, "SAM-assisted remote sensing imagery semantic segmentation with object and boundary constraints," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5636916.

[31] J. Liu, W. Zhu, D. Li, X. Hu, and L. Song, "Domain generalization with semi-supervised learning for people-centric activity recognition," *Sci. China Inf. Sci.*, vol. 68, no. 1, Jan. 2025, Art. no. 112103.

[32] L. Bai, X. Zhang, H. Wang, and S. Du, "Integrating remote sensing with OpenStreetMap data for comprehensive scene understanding through multi-modal self-supervised learning," *Remote Sens. Environ.*, vol. 318, Mar. 2025, Art. no. 114573.

[33] D. Zhang, M. Zeng, and X. Li, "Towards open-vocabulary semantic segmentation for remote sensing images," *Pattern Recognit.*, vol. 175, Jul. 2026, Art. no. 113120.

[34] Y. Zheng, H. Liu, H. Guo, S. Liu, and L. Bruzzone, "SGCA: A scribble-guided cross-attention framework for multimodal remote sensing change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 235, pp. 416–427, May 2026.

[35] P. Zhu and J. Liu, "Joint U-nets with hierarchical graph structure and sparse transformer for hyperspectral image classification," *Expert Syst. Appl.*, vol. 275, May 2025, Art. no. 127046.

[36] P. Xiao et al., "MF-mamba: Multiscale convolution and mamba fusion model for semantic segmentation of remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5405916.

[37] Z. Wang, R. Li, S. Yuan, J. Li, X. Wang, and N. Xu, "Relational structure-aware mamba network for semantic segmentation of remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 3943–3963, 2026.

[38] L. Pan et al., "MambaRSIS: Context-aware multi-scale feature aggregation with selective state space model for remote sensing instance segmentation," *Eng. Appl. Artif. Intell.*, vol. 160, Nov. 2025, Art. no. 111993.

[39] T. Tian, W. Yang, Y. Yin, D. Chen, and X. Tao, "Lightweight UAV image super-resolution method based on large-kernel attention and cross-height strategy," *Expert Syst. Appl.*, vol. 305, Apr. 2026, Art. no. 130693.

[40] M. Zhang, S. Yang, X. Feng, and C. Yaoguang, "Global context-aware multi-scale feature iterative refinement for aviation-road traffic semantic segmentation," *Chin. J. Aeronaut.*, vol. 39, no. 2, Feb. 2026, Art. no. 103693.

[41] X. Pan, X. Xie, J. Yang, X. Song, and G. Shi, "CMANet: Context-aware mutual attention network for referring image segmentation," *IEEE Trans. Multimedia*, vol. 28, pp. 285–296, 2026.

[42] X. Li et al., "Frequency domain-enhanced spectral–spatial fusion transformer for semantic segmentation of remote sensing images," *Inf. Fusion*, vol. 132, Aug. 2026, Art. no. 104248.

[43] X. He, Y. Zhou, B. Liu, J. Zhao, and R. Yao, "Remote sensing image semantic segmentation via class-guided structural interaction and boundary perception," *Expert Syst. Appl.*, vol. 252, Oct. 2024, Art. no. 124019.

[44] K. An, Y. Wang, and L. Chen, "Encouraging the mutual interact between dataset-level and image-level context for semantic segmentation of remote sensing image," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5606116.

[45] C. Zhong et al., "Hierarchical perception guided deep learning framework for landslide detection with multi-source remotely sensed data," *Int. J. Digit. Earth*, vol. 19, no. 1, Jan. 2026, Art. no. 2632415.

[46] J. Li, Y. Huang, X. Ye, and H. Yang, "Topology-joint curvilinear segmentation network using confidence-based Bezier topological representation," *Eng. Appl. Artif. Intell.*, vol. 143, Mar. 2025, Art. no. 110045.

[47] K. Li, G. Vosselman, and M. Y. Yang, "Scale-wise bidirectional alignment network for referring remote sensing image segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 226, pp. 350–363, Aug. 2025.

[48] Y. Chen et al., "Cross-level interaction fusion network-based RGB-T semantic segmentation for distant targets," *Pattern Recognit.*, vol. 161, May 2025, Art. no. 111218.

[49] Y. Gu, S. Chen, X. Sun, J. Ji, Y. Zhou, and R. Ji, "Optical remote sensing image salient object detection via bidirectional cross-attention and attention restoration," *Pattern Recognit.*, vol. 164, Aug. 2025, Art. no. 111478.

[50] Z. Shi, M. Liu, J. Sun, R. Zheng, and S. Dong, "Robust model-based MARL via masked cross-agent completion under observation loss," *Sci. China Inf. Sci.*, vol. 69, no. 4, Apr. 2026, Art. no. 142204.

[51] W. Han, W. Jiang, J. Geng, and W. Miao, "Difference-complementary learning and label reassignment for multimodal semi-supervised semantic segmentation of remote sensing images," *IEEE Trans. Image Process.*, vol. 34, pp. 566–580, 2025.

[52] X. Lu et al., "Uncertainty-aware semi-supervised learning segmentation for remote sensing images," *IEEE Trans. Multimedia*, vol. 27, pp. 5548–5562, 2025.

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[53] X. Li et al., "Frequency-guided denoising network for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5400217.
[54] S. Wang, Z. Shao, D. Hou, and B. Cai, "PV segmenter: A frequency-guided edge-aware network for distributed photovoltaic segmentation in remote sensing imagery," *Appl. Energy*, vol. 393, Sep. 2025, Art. no. 126137.
[55] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, Aug. 2022.
[56] L. Wang, R. Li, C. Duan, C. Zhang, X. Meng, and S. Fang, "A novel transformer based semantic segmentation scheme for fine-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[57] L. Wang, R. Li, D. Wang, C. Duan, T. Wang, and X. Meng, "Transformer meets convolution: A bilateral awareness network for semantic segmentation of very fine resolution urban scene images," *Remote Sens.*, vol. 13, no. 16, p. 3065, Aug. 2021.
[58] X. Liu, Z. Tang, and J. Wei, "Temporal convolutional network based on squeeze-and-excitation module and multi-scale feature fusion for GNSS positioning error correction," *GPS Solutions*, vol. 29, no. 3, Jul. 2025, Art. no. 136.
[59] O. Sharma, D. Trivedi, H. Kumar, S. Pattnaik, and N. Bihari Puhan, "Enhancing intensification forecast of tropical cyclones in the north indian ocean basin using fused vision transformer and convolutional block attention with ResNet model," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4110907.
[60] P. Pan, H. Wang, C. Wang, and C. Nie, "ABC: Attention with bilinear correlation for infrared small target detection," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2023, pp. 2381–2386.
[61] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.
[62] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. 15th Eur. Conf. Comput. Vis. (ECCV)* (LNCS), vol. 11211. Munich, Germany: Springer, Sep. 2018, pp. 3–19.
[63] Y. Si et al., "SCSA: Exploring the synergistic effects between spatial and channel attention," *Neurocomputing*, vol. 634, Jun. 2025, Art. no. 129866.



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